A PROJECT TITLE

ON

Next-Token Probability Explorer for Model Uncertainty Analysis

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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CSE

COMPUTER SCIENCE ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Name of Title: Next-Token Probability Explorer for Model Uncertainty Analysis

1. Abstract:

Large Language Models (LLMs) have achieved remarkable fluency across diverse natural language generation tasks, yet their propensity to produce plausible-sounding hallucinations remains a major obstacle to reliable deployment in high-stakes domains such as healthcare, legal reasoning, financial auditing, and scientific discovery. Standard text generation interfaces obscure the underlying probabilistic mechanics of autoregressive decoding, flattening high-dimensional vocabulary distributions into a single deterministic or sampled sequence. Consequently, vital signals of model hesitation—such as flattened probability distributions, closely competing candidate tokens, and abrupt spikes in predictive entropy—are discarded once a token is selected, leaving researchers and practitioners blind to fragile decision boundaries. To bridge this interpretability gap, we introduce the **Next-Token Probability Explorer**, an interactive visual analytics framework engineered for fine-grained, token-level model uncertainty analysis. The system surfaces step-by-step probability distributions over the vocabulary at every decoding step, enabling users to inspect top-kk candidate tokens,track probability mass concentration, and visualize uncertainty trajectories across generated sequences. By integrating real-time information-theoretic diagnostics—including Shannon entropy, token-level perplexity, log-probabilities, and top-two probability margins—directly onto inline text overlays, the explorer allows users to immediately localize exact token positions where model confidence degrades or diverges. Beyond passive inspection, the framework supports interactive counterfactual branching: practitioners can manually override standard decoding selections at any step by choosing alternative lower-probability tokens, dynamically recomputing downstream semantic trajectories, reasoning chains, and subsequent entropy profiles in real time. Additionally, comparative diagnostic modes facilitate side-by-side evaluation across different decoding hyperparameters (such as temperature and nucleus sampling thresholds), prompt formulations, and model checkpoints. Through targeted case studies spanning factual question answering, multi-step logical reasoning, and syntactically ambiguous prompt resolution, we demonstrate how the Next-Token Probability Explorer uncovers latent failure modes, diagnoses prompt sensitivity, and distinguishes between aleatoric ambiguity in user inputs and epistemic gaps in parametric model knowledge. Ultimately, this tool provides developers, auditors, and researchers with transparent, actionable insights for evaluating generative reliability, systematically refining prompt engineering workflows, and calibrating human trust in large language model outputs.

2. INTRODUCTION:

The rapid adoption of Large Language Models (LLMs) has transformed natural language processing, yet their tendency to generate plausible-sounding hallucinations remains a major obstacle to safe deployment in high-stakes domains such as healthcare, law, and scientific research. A central driver of this vulnerability is the interpretability bottleneck imposed by standard generation interfaces, which flatten a rich autoregressive probabilistic process into a single deterministic text sequence and discard vital signals of model hesitation—such as flattened vocabulary distributions, competing candidate tokens, and spikes in predictive entropy. While existing uncertainty quantification methods typically aggregate confidence at the full-response level, they fail to localize the exact token-level decision boundaries where factual errors or reasoning missteps originate and cascade. To bridge this gap, we introduce the **Next-Token Probability Explorer**, an interactive visual analytics framework designed for granular uncertainty analysis in autoregressive language models. By surfacing step-by-step token probability distributions alongside real-time information-theoretic metrics—including Shannon entropy, perplexity, and top-two probability margins—our system enables practitioners to visually pinpoint epistemic knowledge gaps and aleatoric ambiguity directly within generated text, while further supporting interactive counterfactual branching and multi-configuration comparisons to diagnose failure modes, refine prompts, and calibrate human trust in generative AI.

3. SOFTWARE REQUIREMENTS:

Functional Requirements

- **Model Integration:** Connect to local models (Hugging Face, vLLM) or APIs supporting `logprobs`.
- **Probability Extraction:** Capture top-kk logits and probabilities at every token step.
- **Parameter Controls:** Adjust Temperature, Top-kk, Top-pp, and Max Tokens in real time.
- **Uncertainty Metrics:** Calculate Shannon entropy, perplexity, and top-2 probability margins.
- **Visual Highlights:** Color-code generated text based on token confidence levels.
- **Token Inspector:** Hover/click any token to view top alternative candidates and their probabilities.
- **Counterfactual Branching:** Click an alternative token to force-select it and regenerate downstream text.
- **Data Export:** Export probability distributions and metrics as `JSON` or `CSV`.

Non-Functional Requirements

- **Real-Time Speed:** Stream tokens and metrics with minimal latency (<15 ms overhead).
- **Memory Efficiency:** Store only top 50–100 candidate tokens per step instead of the full vocabulary.
- **Usability:** Clean, responsive web UI with proper subword/whitespace formatting.

Tech Stack

- **Backend:** Python, FastAPI, PyTorch, Hugging Face / vLLM, NumPy
- **Frontend:** React / Next.js, Tailwind CSS, D3.js or Plotly

Signature of the Guide **Course Instructor**